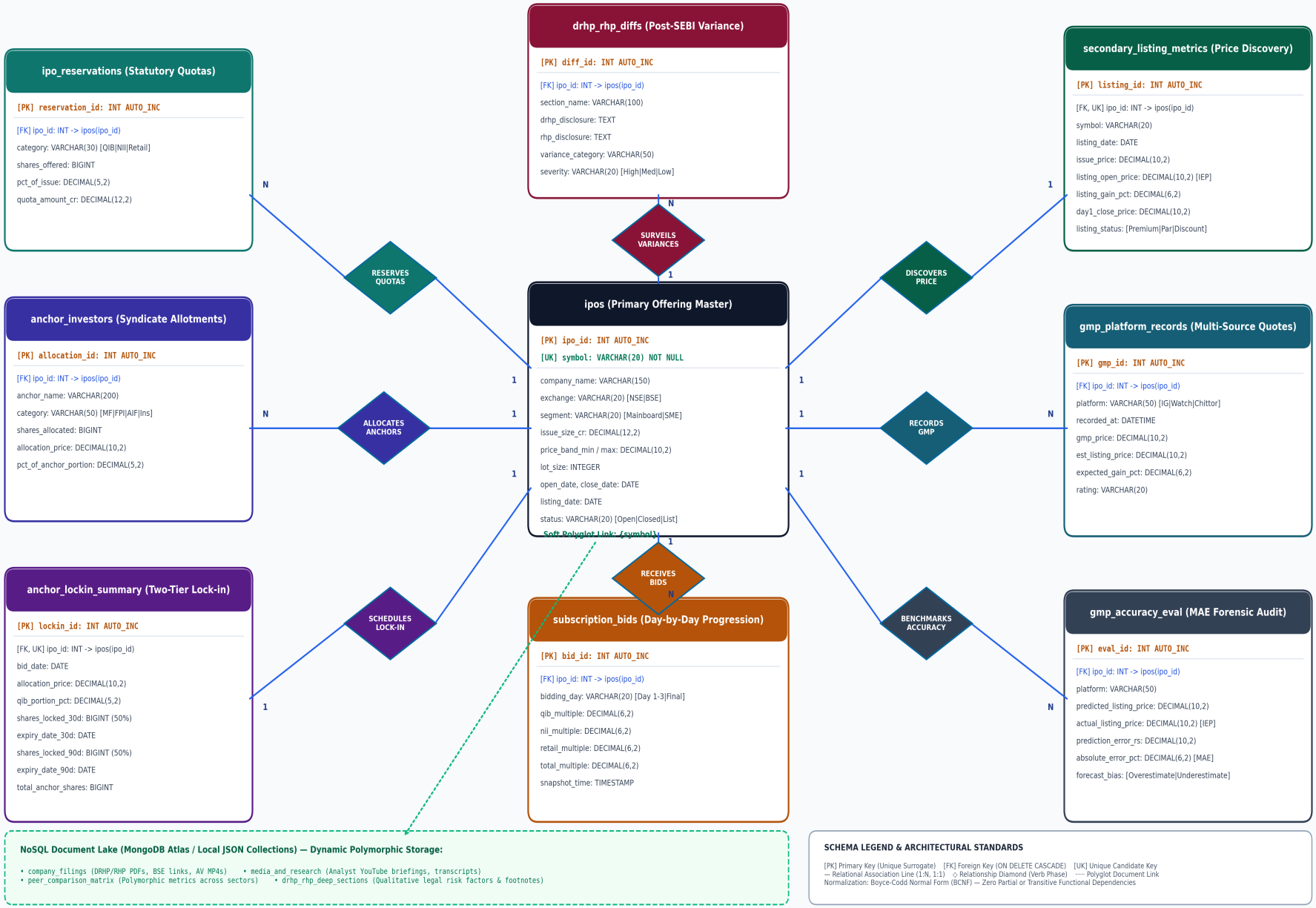
IPOWatch Intelligence Platform (MARK4) — Entity-Relationship (ER) Architecture

Hybrid Polyglot Schema: 9 BCNF Relational Tables + MongoDB Document Lake | Faculty Evaluator: Prof. Smitha G R



Relational Schema Mapping & Formal BCNF Normalization Specification

The conceptual ER diagram above maps directly to **9 Boyce-Codd Normal Form (BCNF) Relational Schemas**. Every functional dependency has a candidate key on the left-hand side. Referential integrity is strictly maintained via foreign keys with ON DELETE CASCADE.

Relational Table	Formal Relational Schema Notation (PK underlined, FK dashed/bold)	Cardinality & Foreign Key Constraint
ipos (Master)	ipo_id, **symbol** [UK], company_name, exchange, segment, issue_size_cr, price_band_min, price_band_max, lot_size, open_date, close_date, listing_date, status	Parent Entity. Primary Key: ipo_id. Candidate Key: symbol.
ipo_reservations	reservation_id, **ipo_id [FK]**, category, shares_offered, pct_of_issue, quota_amount_cr	1 : N from ipos. FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE
anchor_investors	allocation_id, **ipo_id [FK]**, anchor_name, category, shares_allocated, allocation_price, pct_of_anchor_portion	1 : N from ipos. FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE
anchor_lockin_summary	lockin_id, **ipo_id [FK, UK]**, bid_date, allocation_price, qib_portion_pct, shares_locked_30d, expiry_date_30d, shares_locked_90d, expiry_date_90d, total_anchor_shares	1 : 1 from ipos (Unique FK). FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE
subscription_bids	bid_id, **ipo_id [FK]**, bidding_day, qib_multiple, nii_multiple, retail_multiple, total_multiple, snapshot_time	1 : N from ipos. FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE
secondary_listing_metrics	listing_id, **ipo_id [FK, UK]**, symbol, listing_date, issue_price, listing_open_price, listing_gain_pct, day1_close_price, listing_status	1 : 1 from ipos (Unique FK). FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE
gmp_platform_records	gmp_id, **ipo_id [FK]**, platform, recorded_at, gmp_price, est_listing_price, expected_gain_pct, rating	1 : N from ipos. FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE
gmp_accuracy_eval	eval_id, **ipo_id [FK]**, platform, predicted_listing_price, actual_listing_price, prediction_error_rs, absolute_error_pct, forecast_bias	1 : N from ipos. FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE
drhp_rhp_diffs	diff_id, **ipo_id [FK]**, section_name, drhp_disclosure, rhp_disclosure, variance_category, severity	1 : N from ipos. FOREIGN KEY (ipo_id) REFERENCES ipos(ipo_id) ON DELETE CASCADE

Formal Boyce-Codd Normal Form (BCNF) Mathematical Justification

Definition: A relation schema R is in BCNF with respect to a set F of functional dependencies if, for all functional dependencies in F^+ of the form $X \rightarrow Y$, where $X \subseteq R$ and $Y \subseteq R$, at least one of the following holds: (1) $X \rightarrow Y$ is a trivial dependency ($Y \subseteq X$), or (2) X is a **superkey** for R.

Application in MARK4:

• In ipos: Candidate keys are *{ipo_id}* and *{symbol}*. Any FD such as *ipo_id* → *{company_name, price_band_max}* has *ipo_id* as a superkey. There are zero non-superkey determinants.

• **Elimination of Partial Dependencies (1NF → 2NF):** No composite primary keys exist. All non-key attributes depend on the whole surrogate primary key.

• **Elimination of Transitive Dependencies (2NF → 3NF → BCNF):** Anchor investor details, quotas, and bids were factored into independent relations, eliminating dependencies like *ipo_id* → *anchor_name* → *anchor_category*.

Polyglot Relational vs NoSQL Architecture & Viva Voce Defense Guide

This section illustrates how queries are routed between the Relational SQLite/MySQL engine and the MongoDB Document Lake, and equips the student team with definitive answers to faculty ER diagram questions.

Data Entity / Domain	Storage Engine	Structural Schema Justification	Primary Query Pattern in Application
Core Offering Parameters	RDBMS (ipos)	Rigid schema; numeric constraints (CHECK > 0, UNIQUE); BCNF normalized.	`SELECT * FROM ipos WHERE status='Open' ORDER BY open_date DESC;`
Quotas & Anchor Books	RDBMS (ipo_reservations, anchor_investors)	ACID consistency; prevents quota over-allocation (>100%); referential cascade.	`SELECT category, SUM(shares_offered) FROM ipo_reservations GROUP BY category;`
Two-Tier Lock-in Schedule	RDBMS (anchor_lockin_summary)	SEBI statutory 30d/90d split; temporal expiry queries; numerical float calculation.	`SELECT * FROM anchor_lockin_summary WHERE expiry_date_30d >= CURRENT_DATE;`
Statutory Disclosures & Filings	MongoDB (company_filings)	Semi-structured URLs (PDFs, BSE notices, MP4s); avoids empty URL columns in RDBMS.	`db.company_filings.findOne({symbol: 'HYUNDAI'})`
Analyst Video & Media Tree	MongoDB (media_and_research)	Hierarchical arrays of nested video reviews for Google NotebookLM visual tree.	`db.media_and_research.find({symbol: 'SWIGGY'}, {analyst_reviews: 1})`
Cross-Industry Peer Matrix	MongoDB (peer_comparison_matrix)	Polymorphic attributes (Fintech AUM vs Chemical plant capacity) without schema locks.	`db.peer_comparison_matrix.findOne({symbol: 'BLS'})`

Faculty ER Diagram Viva Voce Quick Defense

Faculty: Why do you have a 1:1 relationship between ipos and anchor_lockin_summary instead of putting it inside ipos?
Answer: Under Boyce-Codd Normal Form (BCNF), putting anchor lock-in data directly into ipos would cause NULL value anomalies for offerings that do not have an anchor syndicate (e.g. small SME IPOs that do not utilize anchor allocations). Separating it into anchor_lockin_summary with a unique foreign key ensures that lock-in schedules are only created when an anchor book exists, preventing NULL bloat in the master table.

Faculty: Explain how cardinality is enforced in your SQL database schema.
Answer: 1:N cardinality is enforced by placing the parent's primary key (ipo_id) as a Foreign Key in child tables (ipo_reservations, anchor_investors, subscription_bids) without a unique constraint, allowing multiple child rows per IPO. 1:1 cardinality is enforced by adding a UNIQUE constraint on the foreign key column (e.g. ipo_id INT NOT NULL UNIQUE in secondary_listing_metrics).